SANTANDER X · QUANTUM AI LEAP · 2026

EmpowerTours Quantum Portfolio

Q-Day-resistant DeFi, shipped today.

QAOA on IBM Heron · Hedged post-quantum signatures · On-chain provenance on Monad

THE THREAT

Every ECDSA signature you publish today is a hostage to tomorrow's QPU.

> *Harvest now. Decrypt later. The adversary already has your 2026 trade orders — they just haven't broken the curve yet.*

WHY DEFI GETS HIT FIRST

Public chains are an open archive of signatures.

- Every transaction broadcasts an **ECDSA signature** in the clear.
- Indexers store them forever.
- A cryptographically-relevant QPU forges any one of them.
- The chain doesn't know which.

TODAY'S DEFENSE

None. Most wallets still sign with secp256k1.

Most "post-quantum DeFi" decks are roadmaps.

THE STACK

Three layers. Each one survives the other two.

1. ALLOCATE

QAOA on IBM Heron QPU picks Markowitz-optimal weights under risk constraints.

2. SIGN

Hedged PQ: every order carries ML-DSA-65 + SLH-DSA-SHAKE-256s + Ed25519. Any one survives.

3. ANCHOR

SHA-256(order) committed to Monad's `AuditAnchor` before the vault will execute it.

One layer broken ≠ system broken. The threat model is every algorithm

QAOA on real hardware, reported with statistical honesty.

- Circuit compiled + queued against **IBM Heron** via Qiskit Runtime.
- We report **mean ± stdev across 12 seeds** vs SLSQP baseline.
- Not single-best-run cherry-picking — the failure mode of

WHAT THIS IS, HONESTLY

QAOA is **not** yet faster than SLSQP for this problem size. We measure the **gap** as it closes. Honest measurement is the deliverable.

THE SIGNATURE LAYER

Don't bet on one algorithm. Hedge the bet.

ALGORITHM	FAMILY	WHY INCLUDE
`ML-DSA-65`	Lattice (FIPS 204)	NIST PQ standard. Fast.
`SLH-DSA-SHAKE-256s`	Hash (FIPS 205)	Different math. Slow but conservative.
`Ed25519`	Classical EC	Battle-tested. Hedge for "did we mis-port a new standard?"

> *If ML-DSA falls to a 2028 cryptanalysis paper, SLH-DSA still authenticates the order. If both lattice and hash fall, the classical signature carries it*

THE PROVENANCE LAYER

A hash chain the vault refuses to disobey.

1. Agent builds order. PQ-signs it.
2. `AuditAnchor.anchor(orderHash)` on Monad. Links to caller's previous hash.
3. `RoutingVault.executeAndRoute(orderHash, …)` refuses execution unless `ANCHOR.lastHash[msg.sender] == orderHash`.

WHAT THIS KILLS

- Replay (sequence enforced per wallet)
- Off-chain order tampering (hash doesn't match)
- Vault impersonation (pair

PROOF, NOT PROMISES

Live on Monad MAINNET. 105 tests. ZK-verified.

4
CONTRACTS LIVE
MONADSCAN-VERIFIED

105
TESTS PASSING
PYTHON + FOUNDRY

3
SIGNATURE ALGORITHMS
PER ORDER, HEDGED

AuditAnchor	0x4cb79cc36b367a6fd7363bc6a8553a7a270da27c
UniswapRoutingVault	0xe2fcada067227c817b8a47b850d727ba065e16dd
MorphoSupplyAdapter	0xB1a4341403DA395760561B85C4C96696C0D15958
MLDSAAttestation	0xc1a82D8C4D28Eca8B318D1bac8DCc2Ab963b3839

THE END-TO-END DEMO — REAL VALUE, ONE HASH

QPU decision → PQ signature ZK-verified on-chain → yield.

STEP	CONTRACT	WHAT HAPPENED (MONAD MAINNET)
1. Attest	`MLDSAAttestation`	Order's **ML-DSA-65 signature verified on-chain via ZK proof** (~230k gas, not ~500M); `pqAttested = true`
2. Anchor	`AuditAnchor`	Hash `0xf9e798a1…d3c3` committed, immutable
3. Swap	`UniswapRoutingVault`	0.1 MON → **2,271 USDC** via live Uniswap v3, anchor-gated
4. Yield	`MorphoSupplyAdapter`	USDC supplied into a live **Morpho** market (~4.75% APY), non-custodial

One 32-byte `orderHash` threads all four. A reviewer replays the events

WHY US

We ship the part everyone else handwaves.

- **Most PQ-DeFi pitches:** slides about a future migration.
- **Most QAOA-finance pitches:** one hand-picked seed on a simulator.
- **Most provenance pitches:** off-chain Merkle trees nobody verifies.

- **Ours:** verified contracts, real QPU runs with statistical rigor, on-chain hash chain enforced by the vault itself.
- One repo. One CI. Reproducible from a cold clone.

WHAT'S NEXT

From live mainnet proof to institutional pilot.

Now → Q3 2026

- Independent third-party security audit (contracts + PQ/ZK)
- Package as a B2B SDK (wallets, custodians, bridges)
- HSM/KMS-backed key custody

Q4 2026

- First institutional design-partner pilot
- Hot-path PQ cosigner service
- Paid bug bounty (Immunefi / Code4rena)

THE ASK

Pilot with Santander.

Q-Day-resistant crypto exposure for institutional treasury, on rails we can prove are honest.

`github.com/EmpowerTours/quantum-portfolio` · commit `33b69dd`